

DRAFT | Hybrid recommendation: usefulness, exceptions and exposure

DSAIT4335 Recommender Systems

Group: 24

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Selected development evidence; final test evaluation outstanding

DSAIT4335 Recommender Systems. All reported values are development results, not final test performance or evidence of state-of-the-art performance.

The assignment specifies one page and at most 200 discussion words per task. This draft conservatively allocates one page to each major task (1, 2 and 3).

Development evidence is exploratory. Freeze the reviewed model choices before evaluating held-out tests, and retain the recorded protocol and source hashes with the results.

Group number and actual contributors must be supplied before submission. AI assistance was used; members must review the implementation and record actual contributions.

Method references

[1] Steck (2019). Embarrassingly Shallow Autoencoders for Sparse Data. WWW.
<https://arxiv.org/abs/1905.03375>

[2] Rendle et al. (2009). BPR: Bayesian Personalized Ranking from Implicit Feedback. UAI.
<https://arxiv.org/abs/1205.2618>

[3] Steck (2018). Calibrated Recommendations. RecSys. doi:10.1145/3240323.3240372. Our JSD implementation is an adaptation, not an exact reproduction.

[4] Ning and Karypis (2011). SLIM: Sparse Linear Methods for Top-N Recommender Systems. ICDM. doi:10.1109/ICDM.2011.134.

[5] Kabbur, Ning and Karypis (2013). FISM: Factored Item Similarity Models for Top-N Recommender Systems. KDD. doi:10.1145/2487575.2487589.

[6] He et al. (2020). LightGCN: Simplifying and Powering Graph Convolution Network for Recommendation. <https://arxiv.org/abs/2002.02126>

[7] He et al. (2017). Neural Collaborative Filtering. <https://arxiv.org/abs/1708.05031>

[8] Wang et al. (2019). Neural Graph Collaborative Filtering. <https://arxiv.org/abs/1905.08108>

[9] Aioli (2013). Efficient Top-N Recommendation for Very Large Scale Binary Rated Datasets. RecSys. doi:10.1145/2507157.2507189. Cosine KNN uses the course adapter, not all paper variants.

DRAFT | Task 1: individual models and hybrids

MovieLens 100K; seeds 2026, 2027, 2028. Every observed rating is an implicit positive. RecBole uses per-user random 80/10/10 splits and full-catalog ranking. Previously seen items are masked. Experts train for fixed budgets. Validation users split into 471 meta-fit and 236 development users; 236 users are reserved for policy calibration. This is not a cold-start experiment.

Meta-fit selects expert settings from predeclared grids. EASE [1] and BPR [2] anchor the comparison. Standardized expert scores feed ridge regression, with five sampled unobserved negatives per positive. User, item and disagreement features are ablated. Hybrid penalties (0.001/0.01/0.1/1) are selected on development users. Explicit-rating experts preserve dislikes; their rating-information controls are included.

Constrained weights sum to one but can be negative. The unscaled version operates on z-scores; the calibrated version first learns per-expert affine response alignment on meta-fit users. Effective raw-score coefficients need not sum to one. Outputs remain unbounded ranking scores.

Class models follow [4-9]. FISMCorrected fixes the sigmoid/BCEWithLogitsLoss mismatch and removes each scored item from its own history and normalization. The supplied non-squared norm regularizer is retained: a joint correction, not an exact original-objective reproduction.

All individual models and selected hybrids: mean development nDCG@10

Model	nDCG	Model	nDCG
BPR	0.2204	SignedChannelsLinear	0.2312
ContrastTransfer	0.1396	UserKNN	0.2264
EASE	0.2499	calibrated	0.2612
ExactPop	0.1232	constrained	0.2189
FISMCorrected	0.2220	context	0.2579
GenreContent	0.0244	context-pairwise	0.2376
ItemKNN	0.2192	disagreement	0.2622
LightGCN	0.2263	item	0.2613
NGCF	0.2231	static	0.2620
NeuMF	0.2104	static-pairwise	0.2551
PositiveEASE	0.2294	user	0.2582
Random	0.0071	rrf	0.2519
SLIMElastic	0.2502	group-switch	0.2507

Discussion

Affine alignment matches expert response scales before enforcing sum-to-one weights. The alignment uses sampled zero/one labels; it does not produce calibrated probabilities. Contextual coefficients allow expert contributions to vary with history size, genre entropy and item popularity. Small differences between ablations do not justify a claim that additional complexity improves generalization. The BPR grid changes dimension and training budget together, so their effects cannot be isolated. RRF and activity-group switching provide alternative hybrid controls. Random is excluded from fusion. The best development settings still require a frozen test evaluation; overlapping data seeds are sensitivity checks, not independent replications.

DRAFT | Task 2: effectiveness and interpretation

All metrics are implemented independently: precision, recall, nDCG, MRR, hit rate, catalog coverage, smoothed-frequency novelty, genre Jaccard diversity and genre JSD calibration. Metrics macro-average users with held-out positives. Training-only history terciles and item popularity define groups; head items are the most frequent 20% of the catalog.

Exception study (3 overlapping seeds): likes ≥ 4 , dislikes ≤ 2 , and rating 3 neutral. Seen items are masked. Meta-fit selects settings. Liked-item evaluation excludes users without held-out likes. Its separate cohort and relevance definition are not directly comparable with Task 1.

Separate rating-aware development study: mean nDCG@10 for likes and known-dislike slot rate

Model	Liked nDCG	Dislike rate
Observed EASE	0.2450	0.0162
Contrast transfer	0.1359	0.0060
Signed channels	0.2579	0.0090
Positive EASE	0.2575	0.0085
Pair gate	0.0908	0.0014

Static ridge: mean coefficients on standardized expert scores

Model	Weight	Model	Weight
BPR	+0.0043	LightGCN	+0.0007
ContrastTransfer	+0.0057	NGCF	+0.0080
EASE	+0.0329	NeuMF	+0.0504
ExactPop	+0.0009	PositiveEASE	+0.0178
FISMCorrected	+0.0109	SLIMElastic	+0.0292
GenreContent	+0.0059	SignedChannelsLinear	+0.0021
ItemKNN	+0.0151	UserKNN	-0.0040

Contextual model: mean user and item group results

Group	nDCG	Recall	Diversity	Exposure
Sparse users	0.230	0.329	0.784	n/a
Medium users	0.220	0.236	0.796	n/a
Dense users	0.325	0.152	0.803	n/a
Head items	n/a	0.317	n/a	0.985
Tail items	n/a	0.012	n/a	0.015

Discussion

Static coefficients are associations between correlated standardized scores, not causal credit or probabilities. The group table tests whether aggregate accuracy hides sparse-user or tail-item failures. Contrast transfer achieved 0.1359 liked nDCG versus 0.2579 for separate positive/negative linear channels. The proposed relation mechanism therefore failed this comparison; complexity did not earn its place. The pair-gate revision was motivated by earlier development failures and is exploratory. Information-matched controls separate rating access from architecture. Low known-dislike rates only count observed held-out dislikes: unknown ratings cannot be treated as approval. These findings do not establish emotional understanding, universal superiority or novelty.

DRAFT | Task 3: societal aspects and reranking

Greedy rerankers trade relevance against diversity, calibration [3] or head/tail exposure. The table fixes strength at 0.5, with a top-100 candidate pool. Lower JSD indicates better genre calibration; head exposure is a share, not a fairness verdict. Catalog parity is an explicit diagnostic objective. The ordering comparison uses RRF with constant 60 and retains full ranking support.

UPD is per-user popularity-profile JSD (lower is closer). Discounted catalog exposure Gini changes from 0.939 to 0.927 under item-exposure reranking; lower Gini means less concentrated allocation.

Contextual model: mean development trade-offs

Reranker	nDCG	Diversity	JSD	Head	UPD
Base context	0.258	0.794	0.139	0.985	0.152
Diversity	0.240	0.894	0.125	0.975	0.140
Calibration	0.260	0.804	0.121	0.985	0.152
Item exposure	0.252	0.790	0.140	0.944	0.106
Popularity calibration	0.259	0.792	0.140	0.968	0.130

Before/after RRF: nDCG@10 and objective value (diversity, JSD, head share or UPD)

Objective	Bef. nDCG	Aft. nDCG	Bef. goal	Aft. goal
diversity	0.2576	0.2382	0.828	0.887
calibration	0.2555	0.2529	0.138	0.128
exposure	0.2520	0.2432	0.983	0.957
popularity_calibration	0.2524	0.2524	0.155	0.145

Discussion

Candidate support limits exposure correction: with H head and T tail candidates, a list of k has $\max(0, k-T) \leq \text{heads} \leq \min(k, H)$. Adaptive expansion used 109.4 candidates on average and achieved at least 100% agreement with full-pool exposure reranking in these runs. This measures candidate work, not end-to-end latency. The user-side policy selects exposure strength on independent calibration users with approximate multiplicity-adjusted lower confidence bounds on retention, targeting 95% utility retention. Sparse-group development retention ranged from 99.7% to 100.2%. The observed audit meets the target but is not a future guarantee. Smaller group gaps can hide harm to all groups, so absolute and worst-group utility must also be inspected. Calibration and useful preference exceptions may conflict; that hypothesis needs the separate rating-aware study.